

Machine Learning

Notes of Lect. 1: Introduction

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Università di Pisa - Italy

**Computational Intelligence &
Machine Learning Group**

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Welcome



About ML

- **Machine Learning (ML)**
 - *Master Programme in Computer Science*
 - *Master Programme in Data Science and Business Informatics*
 - *Master Programme in Digital Humanities*
- **Code: 654AA Credits (ECTS): 9 Semester: 1**
- **Lecturer: Alessio Micheli:** micheli@di.unipi.it
- **Material: Moodle** <https://elearning.di.unipi.it/>
 - Self-enrolling mechanism
- **Recordings of autumn 26 lectures: Teams platform**
 - Connect ONLY through the App "Didactic Agenda" ("Agenda Didattica") for 654AA 26/27
 - See also Moodle: «FAQ and general information»
 - Where are the recordings? ML Teams: Shared (tab) → Recordings



Practical information



In class:

- Please, max **silence** during the lecture (to avoid noise): *recording in progress!* (of course, you can ask questions)



Official Decrees: Only in-presence classes (no streaming):

- <All the lessons of the semester will be delivered “**in person**”>, see the general site <https://didattica.di.unipi.it/en/master-programme-in-computer-science/timetable-master-computer-science/>
- Enrolling students' mechanism for attendance “in presence” **and to connect to Teams:** through the App “Didactic Agenda” (“**agenda didattica**”) <https://agendadidattica.unipi.it/>
 - Informally: You can follow the recordings or also the streaming on-line via Teams (but without guaranteed interactions in this case)
 - Note that sometimes recording is *not* possible due to tech. issues

Myself & Aims

- Prof. Alessio Micheli, Dept. of Computer Science, University of Pisa
 - PhD, Full professor of CS, AI, ML
 - Head of [CIML@UniPi](#) (Computational Intelligence & Machine Learning Group): More than 25 years of research in ML/AI
 - *President* of the [European Neural Network Society](#), annual ICANN
 - Past co-chair and co-founder of the [IEEE Task Force on Reservoir Computing](#)
 - National coordinator of the [ML&DM](#) group (including more than 40 research groups) within the [AI*IA](#), annual workshop since 2012
 - Director of the Pisa unit for the [CINI-AIIS National Lab](#) (Artificial Intelligence and Intelligent Systems)
 - Associate Editor of the two main journals in ML/Neural Networks (IEEE TNNLS & Neural Networks - Elsevier)
- **My aim for this course:** to introduce you, in 3 months, to the current ML, also retracing the basics and history

This Lecture

- A. What is ML, and why study ML:**
a motivational introduction
- B. Interlude:** ML in the curricula and other master degrees (not shown)
- C. ML Course info:** prerequisites, course structure, exam, hints, Q&A, bibliography, supports, math background

What is ML

A first look



Prologue: Learning

Learning: universal principles for living beings, society,
andmachines

*The problem of **learning** is arguably
at the very core of the problem of
intelligence, both biological and
artificial*

Poggio, Shelton, *AI Magazine* 1999



What is ML? First view

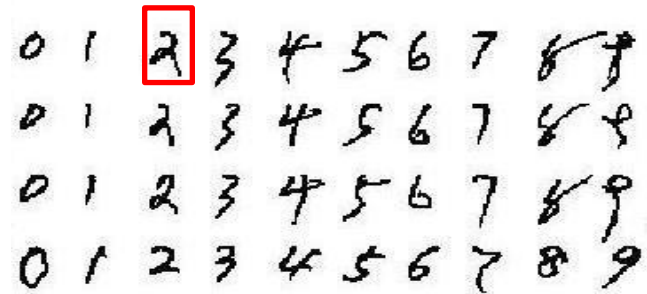
- In Computer Science, a theoretical and applicative field called:
 - Apprendimento Automatico *in italiano (it)*
 - Machine Learning (ML) *English and literature* (aka learning systems)
- **Machine Learning** has emerged as an area of research combining the aims of creating *computers that could learn* (AI) and new *powerful adaptive/statistical tools* with rigorous foundation in computational science
- Machines that *learn* by themselves. Why? Luxury or necessity?
 - Growing availability and need for analysis of empirical data
Central/methodological role due to a change of paradigm in science: data-driven
 - Difficult to provide adaptivity/intelligence by programming (A. Turing)
→ *learning* as the only choice to provide *intelligence* into the systems ...
- **Pave the way to a new AI era** (Growing data+ HPC+ ML flexibility)

What is ML? Simple examples

- Automatized learning by the system of the experience (set of examples) to address a computational task



Email spam classification



Character/face/speech recognition

... no (or poor) prior knowledge/rules for the solution
but it is (more) easy to have a source of
training experience (data with known results)

Zoom: Handwritten Digits

Recognition: Learn a classifier



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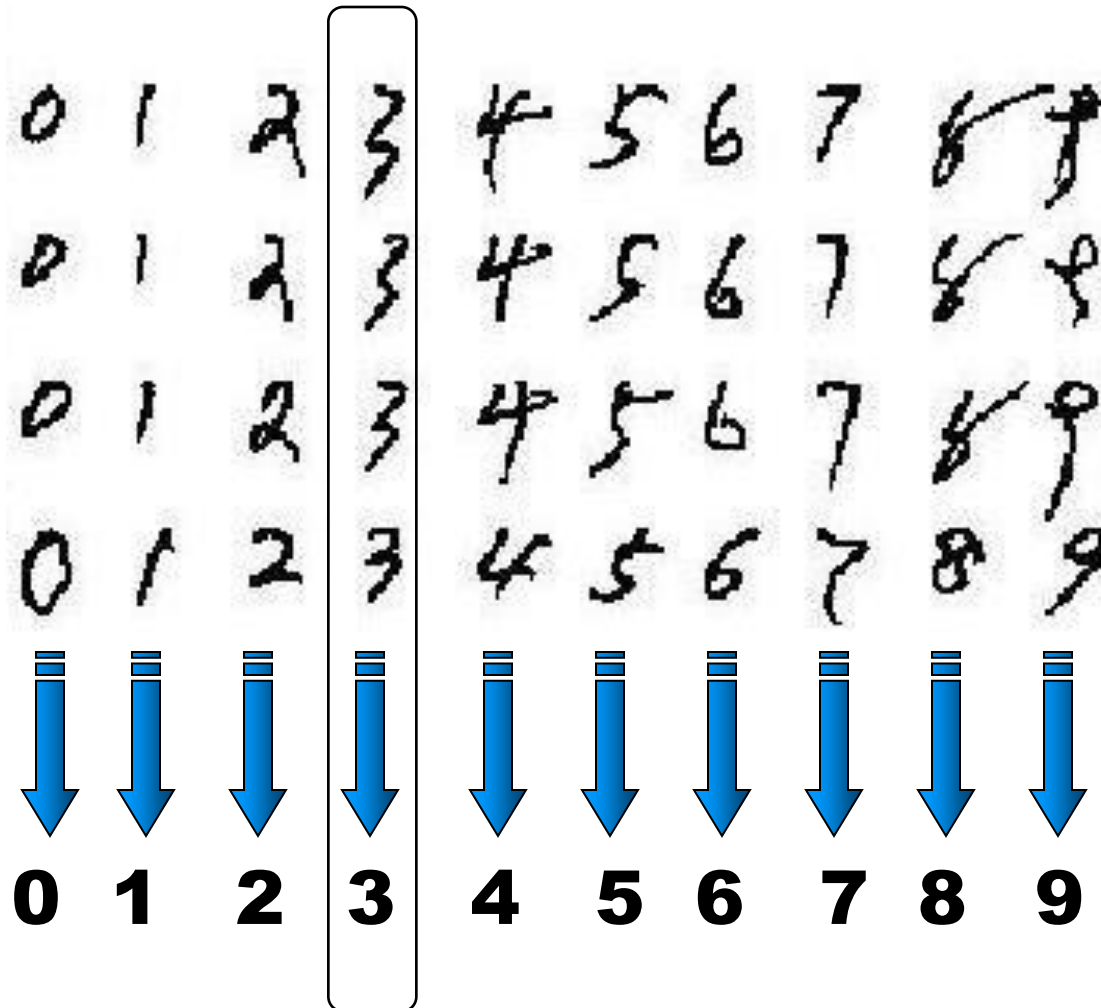
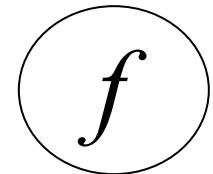
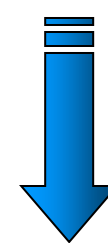


Image
8 x 8



Classifier

Output class

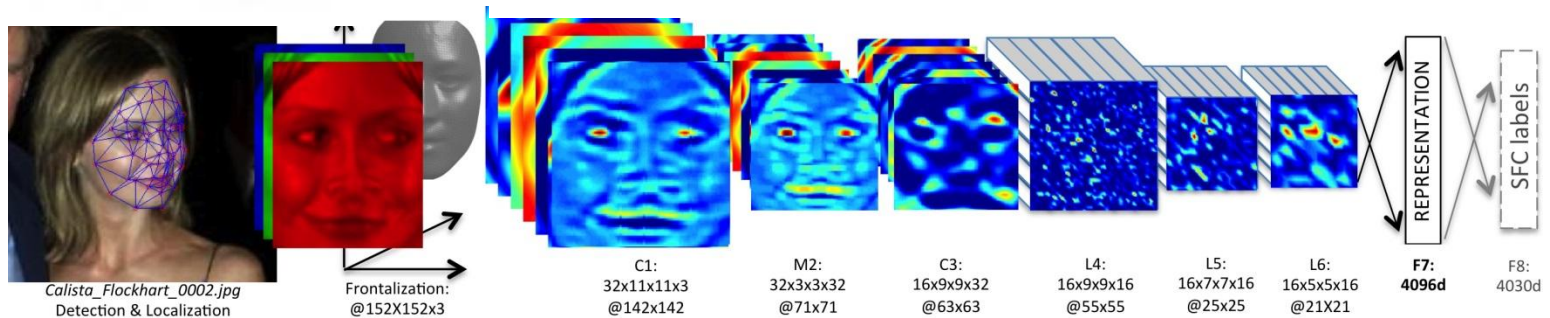
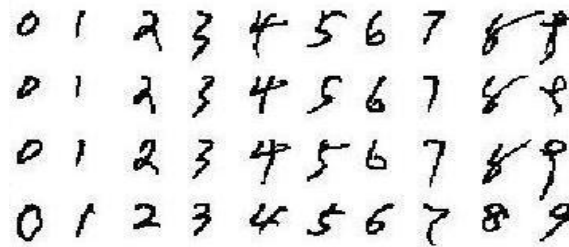


Applicative Areas

Applicative and related areas:

1. Real-World systems (pervasive, from OCR, to human-computer interface, to search engines, ...)
2. New interdisciplinary area, encompassing (since the 80s):
 - **Pattern Recognition** (e.g. face and speech recognition), **Computer Vision**, **Natural Language Processing** (from text, e.g. email spam classification, to speech processing, to generative AI),
 - **Robotics**, Adaptive Systems and Filters, **Intelligent Sensor Networks**, Personalized components, ...
 - **Knowledge Discovery and Data Mining**, **Information Retrieval**, Analysis of complex data (**Med**, **Bio**, Chem, Web, Marketing), Financial forecasting, ...
 - The impact has been revolutionary for some fields, e.g. for NLP, Pattern Recognition and Computer Vision, ... **A "new AI"**

A look by pictures at some current successful applications (SoA)



Face recognition (Facebook)



Go winner
(DeepMind
- Big G)



march 2016



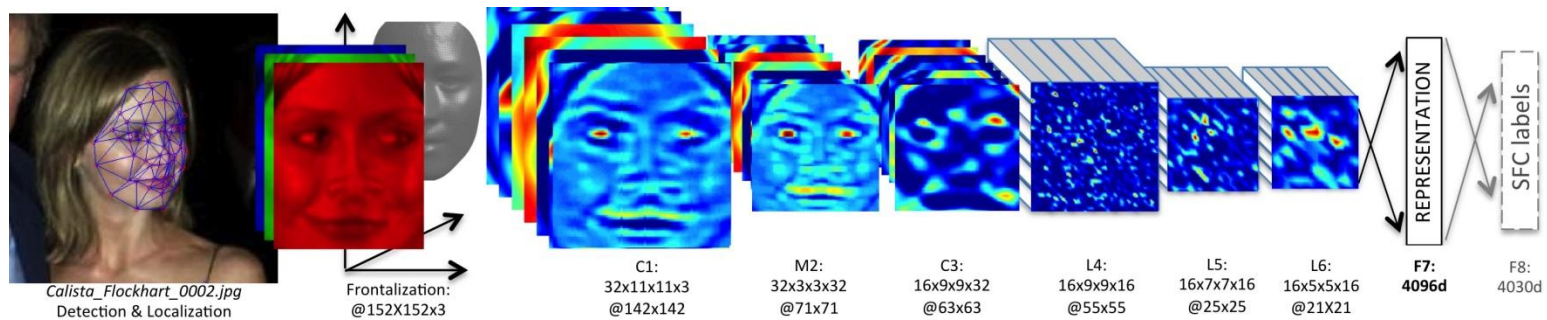
Self-driving cars



Sensor data
Smart *
IoT

An instance of a (not) recent result (CVPR, 2014)

- **Face recognition** combining (deep) **Neural Networks** and other ML approaches
- Starting from four million facial images belonging to more than 4,000 identities



- Asked whether two photos show the same person, DeepFace answers correctly 97.25% of the time ... just a shade behind humans (97.53%).
- From a news in the newspaper LaRepubblica, April 14 ;-)

Health (Examples)

Diagnosis, therapies, personalized medicine, health monitoring... drug design

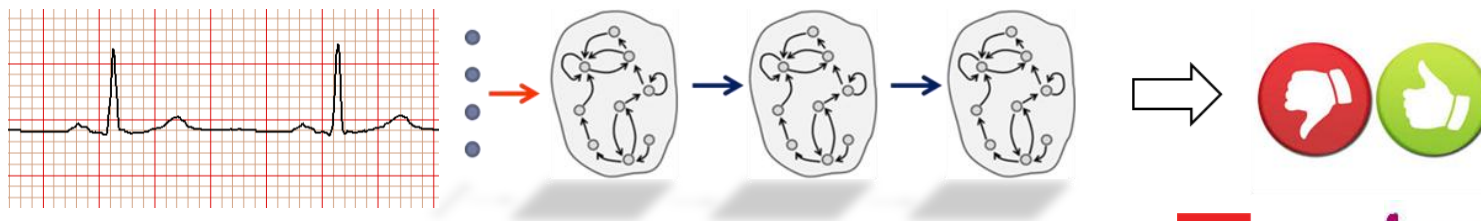
Skin Cancer Classification with Deep Learning



- The system (a deep neural network) can learn from 130,000 cases, far more than a doctor can "in many lifetimes"
- It achieves the accuracy of certified dermatologists (Nature 2017)
- It can be implemented with **app**



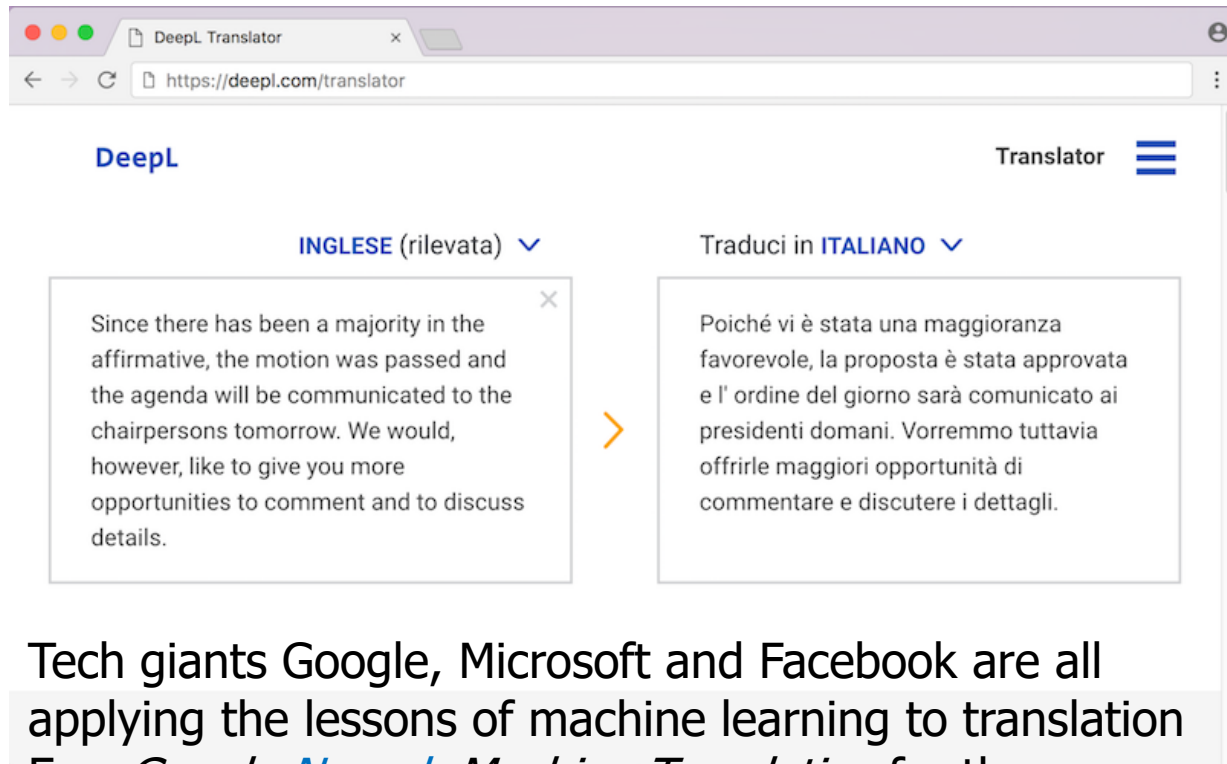
BrAID “Brugada syndrome and Artificial Intelligence applications to Diagnosis”
(PRJ @CIML- UniPi – Tuscany region) [2020- 2024]



Machine Translation



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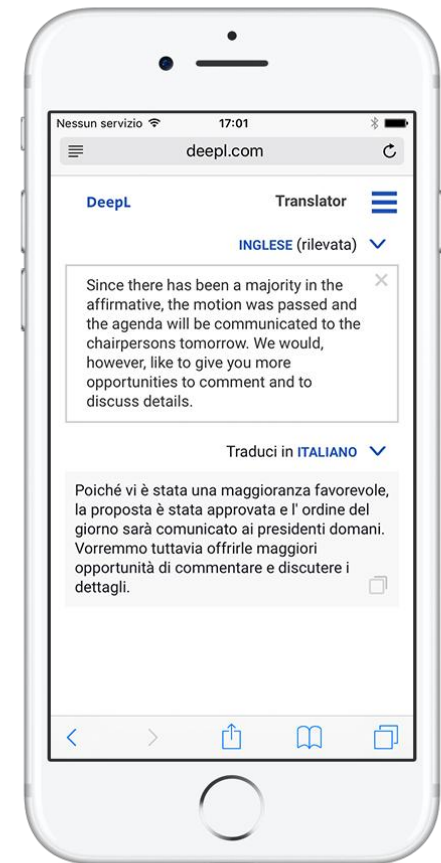
Tech giants Google, Microsoft and Facebook are all applying the lessons of machine learning to translation

E.g. Google *Neural Machine Translation* for the

<Google Translate> tool since 2016, 

but also small companies are making big progresses:

An Example by <DeepL>: since August 2017
a fully *Neural Network* based system.

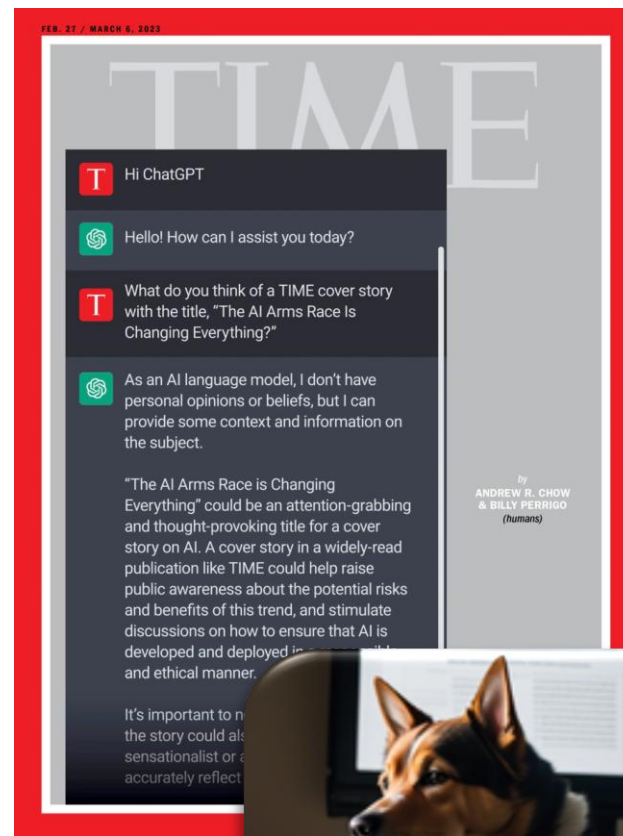


Large Language Models and Generative AI

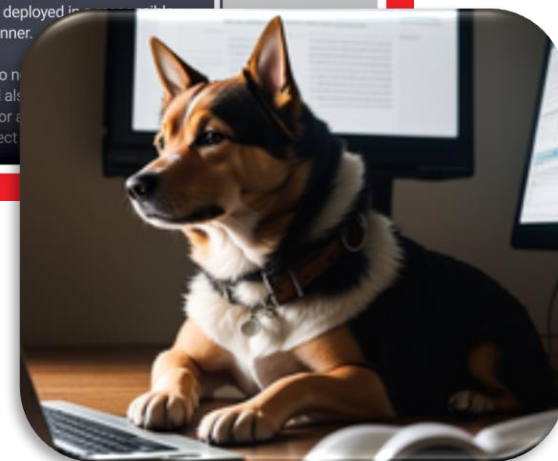


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- Core technology to develop LLMs
 - E.g **ChatGPT**: Chat *Generative Pre-trained Transformer* is a large language model-based chatbot developed by OpenAI and launched on November 30, 2022, notable for enabling users to refine and steer a conversation towards a desired length, format, style, level of detail, and language used.
 - Similarly for other LLM models
 - Dolly x.0 (open source)
- Or systems to generate images, videos, music, etc.



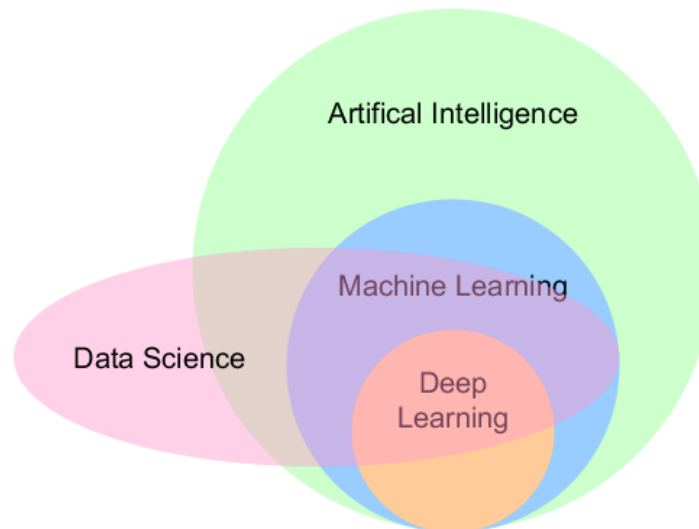
Stable Diffusion, prompt *Cinematic photo of a dog on the Internet editing Wikipedia*



And Related Areas

ML provides the basis (theories and methods) for new AI approaches ([AI revolution](#))

- It opens huge application areas and opportunities
 - Large effort and investments in Industry
 - Latest **GPT** (General-purpose technologies: potential to drastically affect pre-existing economic and social structures)
 - Opportunity to be part of a big challenge for AI/CS/Society
 - “*ML scientist*” (top ranked in job growing demand, e.g. 2019)



*Not the whole story
but a nice view at a
glance*

Ultimate Aim

- Anything that is powerful may be dangerous (by misuses) but the ultimate aim of AI/ML is
 - To bring benefits to people by solving big and small problems
 - To accelerate human progress
 - To empower humans and to add intelligence for any other science field
- AI/ML to *augment our abilities enhancing our humanness in unprecedented ways* (P. Domingo, Scientific American sept. 2018)

HELLO 
BENEFITS

Turing award 2018

- On 27-th March 2019, the Association for Computing Machinery, the world's largest society of computing professionals, announced that Drs. Hinton, LeCun and Bengio had won that year's Turing Award for their work on **neural networks**. The Turing Award, which was introduced in 1966, is often called the *Nobel Prize of computing*, and it includes a \$1 million prize ...
- "For conceptual and engineering breakthroughs that have made deep neural networks a critical component of computing"



Y. Bengio



G. Hinton



Y. LeCun

Nobel Prize – Physics 2024

“for foundational discoveries and inventions that enable machine learning with artificial neural networks”



John Hopfield
Born: 1933, USA



Geoffrey E. Hinton
Born: 1947, UK

ILL. NIKLAS ELMEID © NOBEL PRIZE OUTREACH

- https://www.nobelprize.org/uploads/2024/11/Slideshow_Physics_2024_NobelPrizeLessons.pdf

Nobel Prize – Chemistry 2024

David Baker is awarded "for computational protein design" and Demis Hassabis and John Jumper "for protein structure prediction".



David Baker
Born: 1962, USA



Demis Hassabis
Born: 1976, UK



John Jumper
Born: 1985, USA

- “I’ve always thought if we could build AI in the right way, it could be the ultimate tool to help scientists, help us explore the universe around us.
Demis Hassabis (Alpha Fold - Deep Mind)
- https://www.nobelprize.org/uploads/2024/10/Slideshow_Chemistry_2024_NobelPrizeLessons.pdf

Why study ML?



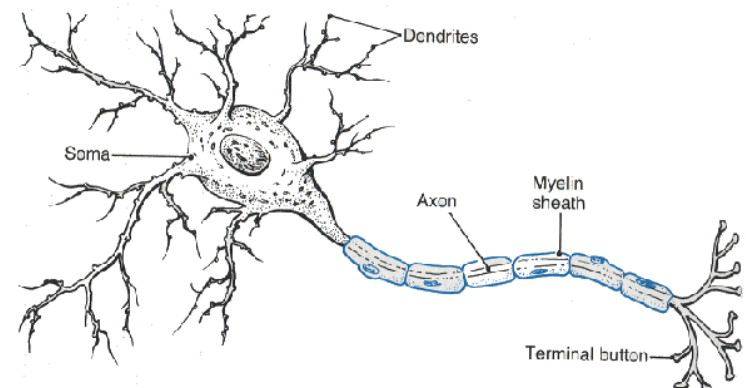
Machine learning in Master Degree



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Why study *Machine Learning* ?

- To **know the basic principles** of learning processes (computational aspects)
- To **know new computing paradigms**
e.g. natural inspired models: **Neural Networks**
 - Studied as computing paradigms since the 40s ' (more than 80 years ago)
 - Neurobiological inspiration
 - Nowadays: set of powerful computing models for *function approximation* and with predictive capabilities supported by a rigorous theoretical ground (*learning theory*)
 - Models for **Deep Learning**
- To be able to **apply** these models with a *rigorous approach*
- **Basic** for all in subsequent AI courses



Modern ML

- From a collection of heuristic methods (often collections of tricks originating from the contributing fields of biology, statistics, neural networks, fuzzy logics, ...) to the construction of *general conceptual frameworks*
- *Fundamental concepts and principles* that govern the learning processes: a new understanding has emerged:
 - Expression of different models in different (Maths/C.S.) “languages” **does not change the principles**
 - Different application fields **do not change the principles**
 - New technological advancements **do not change the principles**
- The core of the *principles* and the *methods* to build flexible/powerful computational models from data is “*the Machine Learning*”

ML aims: points of view

- As AI methodology → [Build Adaptive/Intelligent Systems](#)
 - As statistical learning (inference of hypotheses within math. principles)
→ [Build powerful predictive systems for Intelligent Data Analysis](#)
 - As computer science method for innovative application areas
→ [Using models as a tool for complex \(interdisciplinary\) problems](#)
-
- ❖ All these points of view will be considered in our course,
 - ❖ following a critical approach to discuss the different ML methods with evaluation of their *limitations* and appropriateness of use!
 - ❖ Attention to the principles underlying the methodologies.

Practical aims

Two concrete uses of the course:

- Learn how to *develop new models* in the ML field
- Learn to *apply* current state-of-the-art methodologies for problems in other (interdisciplinary) fields



Great effort: why?

- There are many topics and it is a continuously evolving field: we have to cover them
- ML course to provide solid ground for Data Science and modern AI (and all the following courses)
- For your professional activity, it is expected from **you** to be **the ML expert** !!!
 - Keep it in mind (to motivate your effort)! But also:
 - Interesting: *ML: the driving force of AI* (N. Cesa-Bianchi 2019).
 - A lot of fun: working with learning machines is intriguing
 - A cooperative context for this class will help to work hard and better



Interlude: ML in the curricula

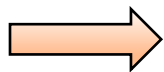
Since 2017 organization by curricula of the CS master programme

PLEASE *read by yourself* the slides: **INTRO-curricula25-v0.1.pdf**

And ask if something is not clear. Thanks.

- Welcome to the new students in CS
- The curricula
- ML in the curricula
- Other master degree students
- Notes for UniPi IIA students
- AA1 (past course) and ML course

PLEASE FILL IN THE FOLLOWING FORM,
help me to know you and to help you:



<https://forms.gle/yZGgV4ddw5oYxeHR7>

***Using the
UNIFI account***

How to study this course



Prerequisites

- No other course is strictly needed (we start from scratch)
- General prerequisites (typically from a First Cycle «Laurea» Degree Programme in CS/Math/Physics/Eng./...):
 - mathematical analysis (functions, differential calculus)
 - elements of matrix notation and calculus
 - elements of probability and statistics
 - algorithms

See a few slides at the end of this lecture for maths background references!



Objectives

- We introduce the principles and the critical analysis of the main paradigms (models and algorithms, with a focus on neural networks) for learning from data.
- Methods:
 - The concepts are progressively introduced *starting from simpler approaches up to the state-of-the-art models* in the general conceptual *framework* of modern machine learning (learning of functions from examples).



The course focuses on the

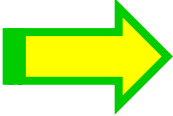
- critical analysis of the characteristics for the design and use of ML
- rigorous experimental evaluation (**use of ML tools *versus* correct/good use of ML**)
- computational aspects of learning systems (CS!)

Course structure - preview

- There is a strong structure in the course:
 - *starting from simpler approaches up to the state-of-the-art models*
 - *the fundamental concepts will be introduced by means of different models*
 - If you follow the *fil rouge* (if you recognize that the main concepts do not change) you gain an easier approach to study the ML course!!!

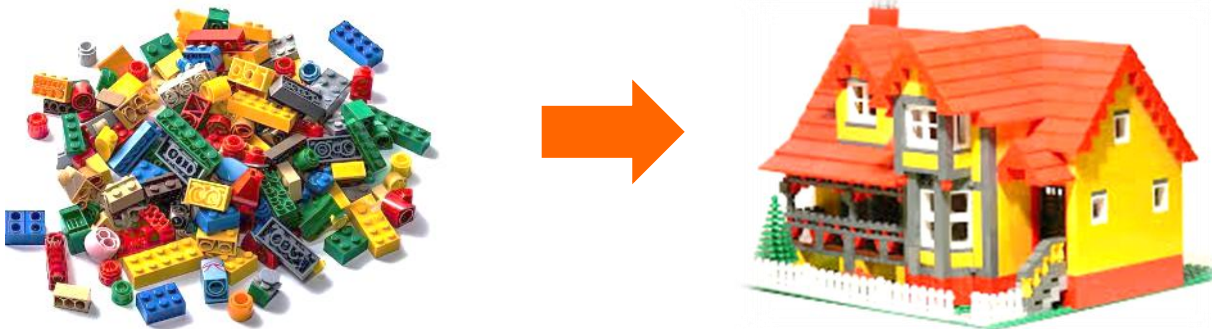
Practical: **1)** Study the single methods
2) Consider the connections/relationships/comparison among them (also to abstract the concepts)
- We will use different “languages” building in parallel different approaches to learn (to infer hypotheses) from data
 - From simpler to more flexible models, along with the support of theoretical results
 - Toward the understanding of the relationship between *flexibility* and *complexity control* as a fundamental concept for the model generalization capability
- A graphical overview will help us to locate the different lectures in the course structure
 - where we are, done, where we are going to, and why




(speech)

Again emphasis on the course structure

- This is NOT just a diagram on the order of the lectures
- It describes our path and the approach: we introduce some «*bricks*» and then will use the **bricks** to build ML models (like NN from linear models or ML systems from validation techniques, etc.).
- In this way things will be simple at the end (by reusing what we did before)
- So keep in mind this: the intermediate models/concepts are often a piece (**bricks**) of a puzzle/building that we will compose later: you have to look back at the end
 - Always follow the structure and please ask for clarification if needed



many bricks to compose our home

Symbol Legend



- Fast in class, often secondary, it will be treated later or useful for your **successive re-reading** (e.g. to connect arguments)



- **Tech argument**, fast in class, *important topic* but **not** strictly need in order to follow the course main line at a first glance, but **useful** for your successive deeper reading (e.g. because it is easier to be understood *later*, or useful to develop the *project*, etc.)

Exercise

- "*Exercise*": something *useful* **to do** by yourself

Course contents (skip now)

(Indicative synthetic programme)



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WARNING: See the final program provided at the end of the course

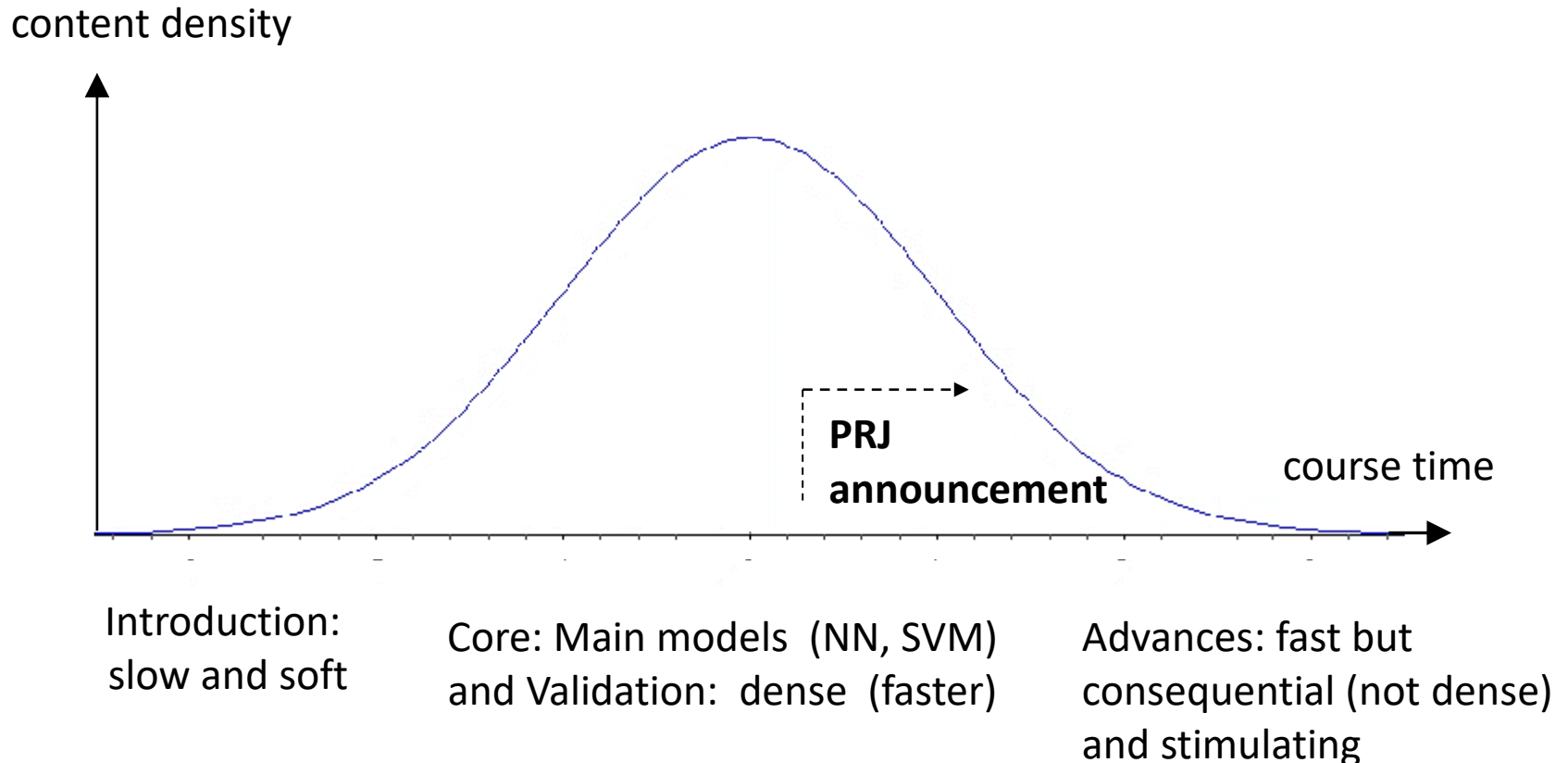
- **Introduction:** Computational learning tasks, prediction, discrete and continuous spaces, inductive bias, generalization
- **Basic concepts and models:** linear models (models and properties), nearest neighbor (models and properties), propositional and rule based models.
- **Neural Networks (NN):** Perceptron and computational properties. Introduction to multilayer feedforward NN: architectures and learning algorithms*. Regularization. Recurrent NN*. *Intro to recent NN paradigms**: Deep Learning, Randomized NN.
- **Principles of learning processes and general practical aspects:** Validation: model selection and model assessment, Bias-Variance analysis. Elements of Statistical Learning Theory, VC-dimension.
- **Support Vector Machines:** linear case, kernel-based models.
- **Bayesian and Probabilistic/Graphical models** [removed]
- **Unsupervised learning:** vector quantization, self-organizing maps.
- **Introduction to Benchmarks and Applications ***
- **Advanced approaches:** Learning for Structured data (intro)*

*** These parts have been progressively extended**

Course Intensity/Speed Distribution through Time



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Assessment Methods

Exam



Assessment Methods

Guide



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In this chronological order

1. Exercise Tests during the course (optional): self-assessment, witness of your active participation
2. Optional project (at the end of the course, December)
3. **Written exam** (after the course, at sessions time, starting on January and February)
4. Oral (only if need) and exam finalization

Details in the following

1. Exercise Tests



- Short and very simple **intermediate exercise tests** will be provided during the course, at the end of each cycle of lectures, through **Moodle**
 - They are provided for your self-evaluation, *as homework exercises*, with a score, but it is *not* a score valid for the exam (self-evaluation aim)
 - They are available **only** for periodic exercises *during the course*
 - They are strongly suggested to *keep up with the lessons (up to date)*, but not mandatory (see also later)
 - They can anyway show me your *active presence* during the course: they have a closing time, typically after 1 week
 - But you can continue the series if you miss one or some of them!
 - They allow you to be *better prepared* for the Written Test (even if the format, the content and the depth differ)
 - The Exercise tests will be **announced** *only* in class and *through Teams* by a short message

2. (Optional) Project



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- Students have the **opportunity** to develop a project
- **Optional** (this year the prj is **NOT mandatory**)
 - Team work: It is made by a **team** of 2 students: no exceptions
 - Content: Implementation, application and/or analysis of a *specific* ML/NN method/technique with experimental evaluation. Possible coordinate prj with CM*
 - Format: a **poster** public presentation in a classroom (+ developed material delivery)
 - Timing: **only** at the end of the course (one of the last lessons)
 - Suggestion: do not apply for a project before to fully **study the course content**
 - Anyway, after the project presentation you can do the exam in any session (even if it suggested not too late, see before)
 - Best poster award 
- Of course, in order to discuss the project, the members of the team need to do the **exam in the same session**

Details will be provided in a specific lecture later

3. Written exam (I)



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- Remember that it is **mandatory to register** for the exam on *esami.unipi.it* portal till the deadline reported in that site.
 - Check the **deadline** *well in advance*
 - *Only* registered students can be to the exam (all students rule)
 - Registration must be canceled if you decide not to participate to the exam
- The written test will be held in person only (**in presence**).
- The test will take place in a **classroom** on the **date**, at the time, and in the room(s) specified for the exam session on *esami.unipi.it*.
- **The written test is mandatory.** *Failure to attend the test at the scheduled date, time, and place for the exam session will be considered as withdrawal from that exam session.*

3. Written exam (II)



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READ THIS SLIDE BEFORE THE WRITTEN EXAM

- There will be questions to answer on pre-printed **paper sheets** with a space provided for a free-form responses
- You must bring a **pencil** or **pen** to write your answers on paper
- Written questions require answers using **equations** and/or drawings and/or descriptive parts (quiz and open questions)
- **No** course or other external **material** may be used during the test.
- The use of any **electronic aid** (virtual assistants, chatbots, etc.) is *not allowed* and will result in exclusion from the test
- In particular, the use of **devices** during the test is strictly prohibited, including laptops, smartphones, smart bracelets, earphones, smart glasses and similar devices.

4. Oral and exam finalization

Possible **oral and recording of the exam grade** (it. “verbalizzazione”):

- To access the oral exam, students *must have obtained a **sufficient** level in the written tests*
 - list of *admitted* will be communicated through Moodle
- Oral is the **same session** of the written exam
- The oral exam finalizations will be made group by group ***in presence***, typically in my office at the Dept. of Computer Science.
- *What is an oral? **If need** (depending on the written exam result) an oral exam consists of an interview between the candidate and the teacher on *the whole course content* and, if useful, includes a *discussion of the written test* and of the *optional project developed*.*

4. Oral and exam finalization Timing



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- In the next days after evaluating and marking the written test, I will set up a **slot** for you/your group, and I'll communicate it to you.
- Of course, the orals need to be distributed **across different days** starting after the Written Exam, i.e. as soon as possible in one of the following days/weeks
- Special cases:
 - *In case of project*, the oral exam finalizations will be made group by group: (in order to discuss the project together), **i.e.** both the teammates need to do the exam in the same session
 - *Specific needs* (e.g. early oral priority, problem to be in presence, etc) will be discussed case by case upon request. I'll try to help you if the overall scheduling of requests allows me to do it.



Exam: **Summary** of Rules (I)

Exam evaluation:

- **Written results**
 - + active participation (including test exercises during the course)
 - + extra (not mandatory) activities (a possible prj, a volunteer presentation at the end of the course, ...),
 - + full oral, only if requested by the teacher
- **If written result is sufficient** → at a given slot some days after the written exam (I'll communicate it to you) → possible oral and mark proposal → if accepted: final recording of the exam grade
- **If written result is not sufficient** or mark not accepted → next session (at your choice among the many we have during the year, according to Unipi rules and calendar)



Exam: **Summary** of Rules (II)

- Online **exercise tests** during the course are useful for self-assessment and help students to be better prepared for the exam
 - Attending the course and completing (80% of) all online exercise tests can be also a small plus for the exam (if the written part was sufficient)
 - However, it is NOT necessary for taking the exam, nor is it required to achieve a top grade
- The **optional project** may be an interesting experience for yourself
 - A very good project is also a small plus for the exam (if it was already good)
 - However, it is NOT necessary for taking the exam, nor is it required to achieve a top grade
- The **written and oral examinations** will assess all these aspects

Your final knowledge of machine learning, and the depth of that knowledge, are the key performance indicators for the final exam grade

The main hints (I)

- Follow the lectures and slides as a guide, studying *progressively* during the course
 - Special interactive classes will be used to assess activities and make a ML discussion forum (**Q&A classes**). This is a class, not a set of records*.
 - Within a **cooperative** and collaborative context!
 - *Repetita*: The **intermediate tests** are meant to stimulate you to **stay "online/ up to date"** and to have constant **self-evaluations**, hence useful for you!

The main hints (II): Help!

If something is *not* clear to you:

- Don't worry
- Ask me → see later *Q&A* modality
- Collaborate with your teammates/colleagues
- Use the books/papers : *bibliography* + personal search → see later for bib
- Use the course supports → see later
- Study, **it's ML!!!** It's a constantly evolving field, ...
 - go back to the slide "Great effort: why?"

Other practical info

- **Lectures schedule:** (starting time is sharp to have a break)
 - Tuesday 16-18, Aula E
 - Wednesday 16-18, Aula E
 - Thursday 16-18, Aula E
- **Q&A (with the teacher):** Tuesday **after lecture***+ *special Q&A classes*.
 - * NOTE: the time after the lecture IS NOT mandatory, it is not part of the lecture, it is for single/ *collective* student Q&A reception
 - Or write to me an email for very personal questions, using the TAG **[ML-26-Q]** in the subject (**the tag is a must!**)
- **ML mailing list:** I'll use Teams and/or Moodle news
 - (take care to have on it the email address that you really use)
- **Make up for lost lectures:**
 - Monday afternoon (they will be announced later, on the need)
 - Lesson recovery **09/12 – 18/12/2026** by academic calendar (consider this period before programming your Christmas vacations travels!)*

Last News



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Announcement

- **No ML lectures on 15-17 September 2026**
- We will recover these 3 lectures later

Textbook and material (1)

- **Course notes (slides copy)**

with material from time to time indicated in the classroom:
bibliographic references at the end of each specific topic via
book chapters and/or online material.

Hints:

1. *Course notes are a very useful **guide** to the selected topics!*
2. *Have a look **in advance** (I typically provide them in advance on Moodle)*
3. *Take **your notes** [past student suggestion]: make connections among slide phrases, and slides themselves can be not sufficient to reconstruct the lecture line of thought*
4. *I would like to share on our Moodle your notes of the course: please, contact me if you like to **share your notes** (in electronic form)*
5. *Learn to take advantage consulting **different sources** (**master degree level**): see next page for the books and the references in the slides*



How to get the slides?

The electronic version of the slides is intended only for authorized students, **for their personal use.**

- The link to the electronic slides is reserved (draft material). **It is not a public site!**
- **Material: Moodle:** <https://elearning.di.unipi.it/>
 - See **Machine Learning 2026**
 - See also the **FaQ and general information** section therein
- Please do not use the link in any web site/social media
- Please do not repost slides in any form
- Take care of the last version file ML-26...**-v*** (sometime after the lecture!!!)



How to get the slides? (II)



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- If you not have a **UNIFI student account** yet:
- The didactics offices typically provide the possibility to connect to the Unifi telematic services (Moodle, Teams, etc ...), your credentials, even before formally enrolling, please inquire by yourself
- Try connecting to the [ALICE service](#)
 - If really nothing works, please ask me by email for a TEMPORARY guest account *for Moodle*

TEMPORARY

Textbook and material (2): Books



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- **General Bibliography** (library or on-line):
 - S. Haykin: ***Neural Networks: a comprehensive foundation***, IEEE Press; (2nd. Edition, 1998) → OR better
S. Haykin: ***Neural Networks and Learning Machines***, Prentice Hall; (3rd Edition, 2008)
 - T. M. Mitchell: ***Machine learning***, McGraw-Hill, 1997
 - I. Goodfellow, Y. Bengio, A. Courville: ***Deep Learning***, MIT Press, 2016
On-line: <http://www.deeplearningbook.org/>

Textbook and material (3)

- **Other references (examples):**
 - (AIMA) Russell, Norvig: *Artificial Intelligence: A Modern Approach*
 - Hastie, Tibshirani, Friedman: ***The Elements of Statistical Learning***, Springer Verlag, 2001 (new eds. up to 2017) → 2017 [Free on-line copy](#)
 - Cherkassky, Mulier: *Learning from data: concepts, theory, and methods*, Wiley, 1998 (new: 2nd Ed., 2007)
 - C.M. Bishop: *Pattern Recognition and Machine Learning*, Springer 2006 – freely available from the web site
 - Bishop: *Neural Networks for Pattern Recognition*, 1995
 - Duda, Hart, Stork: *Pattern Classification*, 2nd. ed. J. Wiley & Sons, 2001
 - Shalev-Shwartz, Ben-David: ***Understanding Machine Learning: From Theory to Algorithms***, Cambridge University Press, 2014, [Free online copy](#)

Many technical books/blogs exist, take care of the scientific quality!

Textbook and material (4)

For software libraries:

- Books that describe *Keras / Scikit-Learn*
 - "Deep Learning with Python" (F. Chollet)
 - "Hands-on Machine Learning with Scikit-Learn, Keras & Tensorflow" (A. Geron).
- Using PyTorch, NumPy/MXNet, JAX, and TensorFlow

```
@d2l.add_to_class(LinearRegressionScratch)
def loss(self, y_hat, y):
    l = (y_hat - y) ** 2 / 2
    return l.mean()
```

 - "Dive into Deep Learning" (A. Zhang, Z. Lipton, M. Li, and A. Smola) (2023). Cambridge University Press. Freely available from the website.
- We will have an introduction to the main libraries later (by the course assistant)

Many technical books/blogs exist, take care of the scientific quality!

Supports

- This year we will have a ML course assistant to help you for the *project development (TO BE CONFIRMED!)*
- Take the opportunity to interact with her/him during the course



Other Supports: ML Chatbot



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- A ChatGPT tutor has been created on an experimental basis to assist with studying for the ML 2026 course
- You can use it to clarify doubts, explore concepts in greater depth and get guidance on the slides, definitions and exercises, always based on the official course materials
- Details and the link will provide later (news of Moodle)

For students who arrived late

- **All the information** for the course are in the **Moodle** of the Machine Learning course, entering into *elearning.di.unipi.it* using the Unipi account (Self-enrolling mechanism)
- Carefully read **lecture slides**, and bibliography references when need: all the material is in the Moodle
- Use also:
 - Teams for news and recordings
 - Notes by your colleagues (I suggest to contact them soon)
 - Keep in mind that the project and the exercise tests during the course are not mandatory for the final exam.
 - *FAQ and general information* section in the Moodle
 - You must use the TAG "[ML-26-Q]" in the subject for possible other questions (if any).
 - *Don't ask* the teacher what is already reported in the *FAQ* section or in the first lecture.

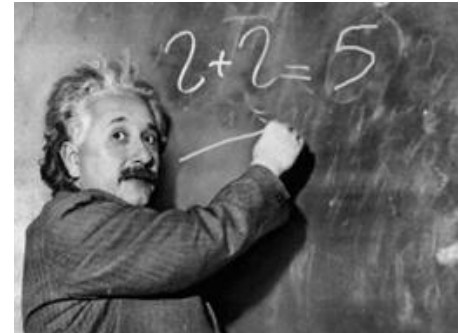
Background



Background 1

Very informal maths preliminaries:

- E.g., AIMA appendix A.2 and A.3 (mathematical background) freely available at <http://aima.cs.berkeley.edu/newchapa.pdf>
- Or Deep Learning book chapters I.2, I.3, I.4 (i.e. part I: 2,3,4) <http://www.deeplearningbook.org/>
- Or (much more) *Mathematics for Machine Learning* (2020) <https://mml-book.github.io/>
- And feel free to ask for any doubt!!!!
- Don't worry: AI/ML are multidisciplinary fields from the beginning!!!!



Background 2

NEW since 2017:

- In parallel we have the course of CM (*Computational mathematics for learning and data analysis*) or *Optimization for Data Science* (@DSBI)
- Great synergy with ML
- Its nature and aim is to provide a deep mathematical background for learning methods. You can consider such lectures
 - Timetable: please, check by yourself
- Anyway, the course is *not* mandatory to follow ML, as many students in the past from other Master Degrees or curricula without CM (I'm aware of this)

Notation

Generally, we will use consistent notation among the most important parts. However:

- Sometimes *locally defined* notation will be used to *simplify* the concepts introduction and/or only if needed to be consistent with *different books*
 - Anyway, the equivalent symbols for the same concept will be indicated
- A document with the “**notation summary**” of the course will be provided (after the models discussion, to resume).
- In any case, don’t go crazy for notation issues: please ask me if you have some issues, I will be happy to help (better immediately than later) and to correct if needed

Basic background (references)

- Multivariable/Multivariate calculus
Functions with multiple inputs: $f(x_1, x_2)$ or $f(\mathbf{x})$, where $\mathbf{x} = (x_1, x_2, \dots)$
 - *Partial derivative, gradient.*
- Probability:
 - *probability density function, mean and variance, Normal random variable*
 - *$p(x)$ or $P(X=x)$, conditional probability: $p(x|y)$, joint probabilities, ...*
- Matrix calculations and notations: x (scalar), $\mathbf{x}$ (vector), $\mathbf{X}$ (matrix)
 - *inner (dot, scalar) product, inverse, norms, ...*
- *Notions that can be studied in parallel (e.g. Numerical Methods and Optimization courses and/or the mentioned CM course):*
 - *The linear least-squares problem: use of the Singular Value Decomposition*
 - *Optimization: Lagrange multipliers*

(it) *Prodotto Scalare*

(en) Inner (dot, scalar) product

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = \sum_{i=1}^n a_i b_i \quad \mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$$

- Other notations $\mathbf{a} \cdot \mathbf{b} = \mathbf{a}^T \mathbf{b} = \mathbf{a}^t \mathbf{b} = \langle \mathbf{a}, \mathbf{b} \rangle$ and even $\mathbf{ab}$ if the context is clear
- Magnitude/size (it '*Modulo*') or length or *Euclidean norm* of a vector $\mathbf{x}$

$$\sqrt{\mathbf{x}^T \mathbf{x}} = \sqrt{\mathbf{x}^T \cdot \mathbf{x}} = \sqrt{\sum_i x_i^2} = d(\mathbf{x}, 0) = \|\mathbf{x}\|_2 = \|\mathbf{x}\| = |\mathbf{x}|$$

Simplified

- Generalization: function that relates a couple of vectors to a number (a scalar): (bilinear symmetric form, by the notation $\langle, \rangle$)

$$\langle v, w \rangle = \langle w, v \rangle$$

$$\langle v + w, u \rangle = \langle v, u \rangle + \langle w, u \rangle$$

$$\langle kv, w \rangle = k \langle v, w \rangle$$

- Cauchy-Schwarz inequality $|\langle x, y \rangle| \leq \|x\| \cdot \|y\| \quad \forall x, y \in V$

Tensors & other defs.

Tensor

- Simple view: an array of numbers arranged on a regular grid with a variable number of axes is known as a tensor.
- i.e. A **multi-dimensional array** of numerical values.
- E.g. $X_{i,j,k}$ (3 indices)
- Used trivially in ML just for notational advantage, but in general with more mathematical properties useful when dealing with changes of basis, basis independence etc. in linear algebra, geometry, physics
- In ML they help focus when using GPU etc. (as we will see later)

Other defs.

$$\|\mathbf{x}\|_1 = \sum_i |x_i| \quad L^1 \text{ norm}$$

$$\|\mathbf{x}\|_\infty = \max_i |x_i| \quad \text{Max norm}$$

Partial derivative: calculus

- The **partial derivative** generalizes the notion of the derivative to higher dimensions.
A partial derivative of a multivariable function is a derivative with respect to one variable (e.g. x_1) with all other variables held constant.
- Hence, if $Df(x) = f'(x) = df/dx$
- for $f(x_1, x_2, x_3)$ we can compute: $df/dx_1, df/dx_2, df/dx_3$
$$\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2} \quad \frac{\partial f}{\partial x_3}$$

Gradient

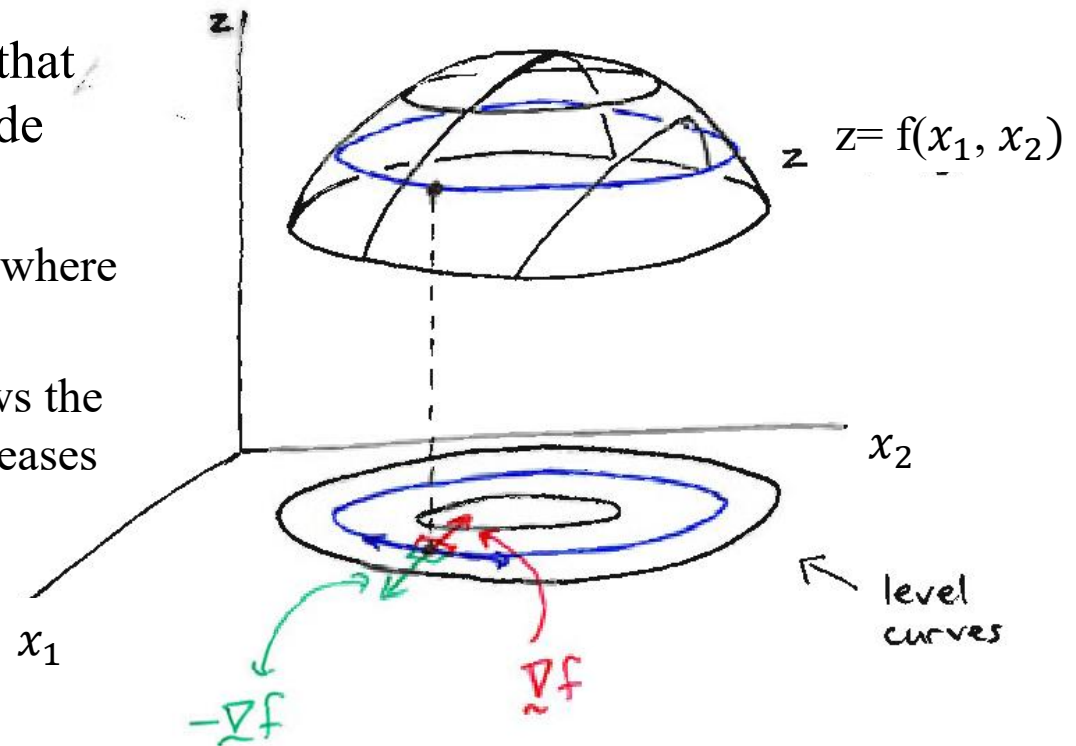
- When a function of two variables $f(x_1, x_2)$ has partial derivatives at each point (x_1, x_2) we can associate the **vector** of the two partial derivatives $\left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}\right)$, called the **gradient** of f , often denoted as ∇f or *grad f*.
- For n variables: the gradient is the vector field whose components are the partial derivatives of f :

$$\text{grad } f = \mathbf{e}_1 \frac{\partial f(x_1, \dots, x_n)}{\partial x_1} + \dots + \mathbf{e}_n \frac{\partial f(x_1, \dots, x_n)}{\partial x_n}$$

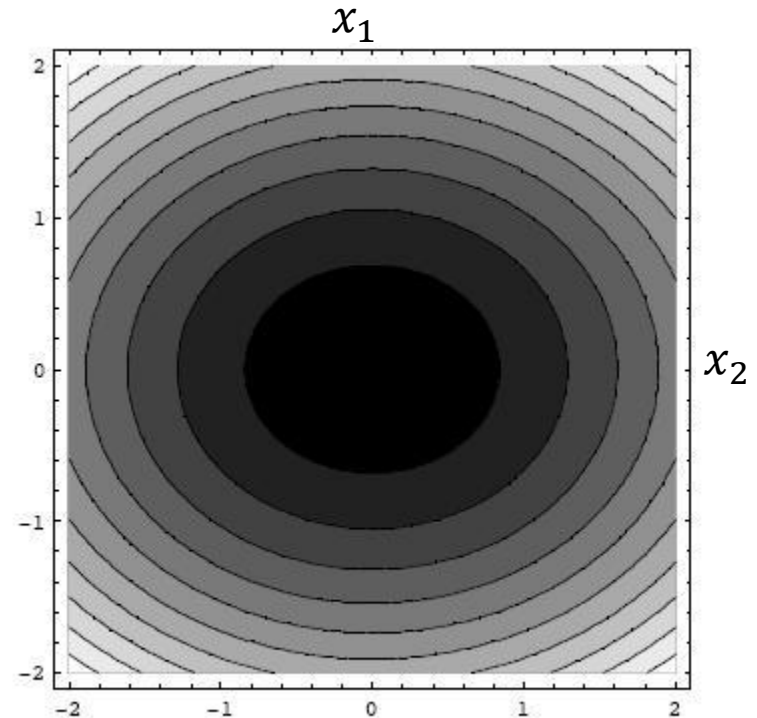
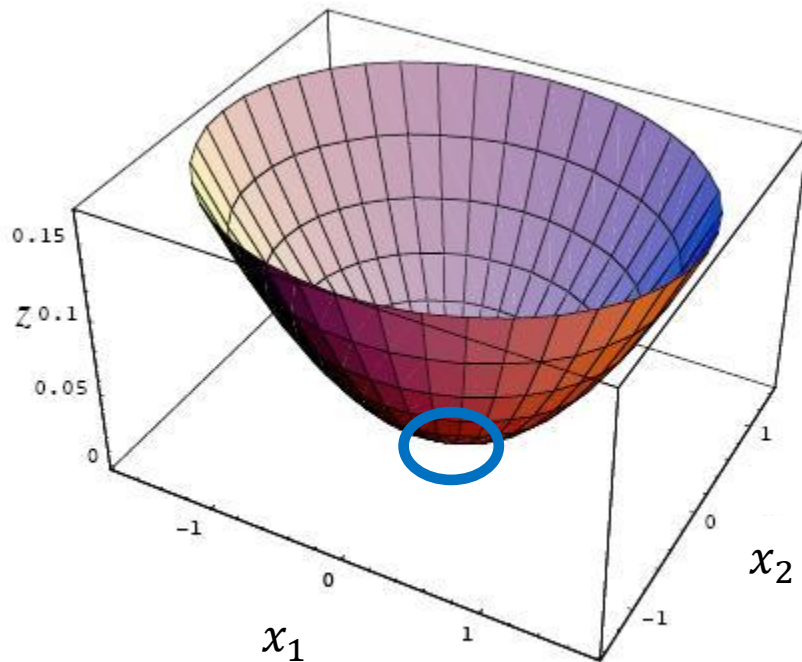
where the $\mathbf{e}_i$ are the orthogonal unit vectors pointing in the coordinate directions (*e.g.* $001, 010, 100$ in 3D).

Gradient on a surface

- The gradient at a point is a vector pointing in the **direction of the steepest slope** at that point.
- The steepness of the slope at that point is given by the magnitude of the gradient vector
 - The **gradient** shows the direction where the function grows
 - The **negative of the gradient** shows the direction where the function decreases



Local Minimum

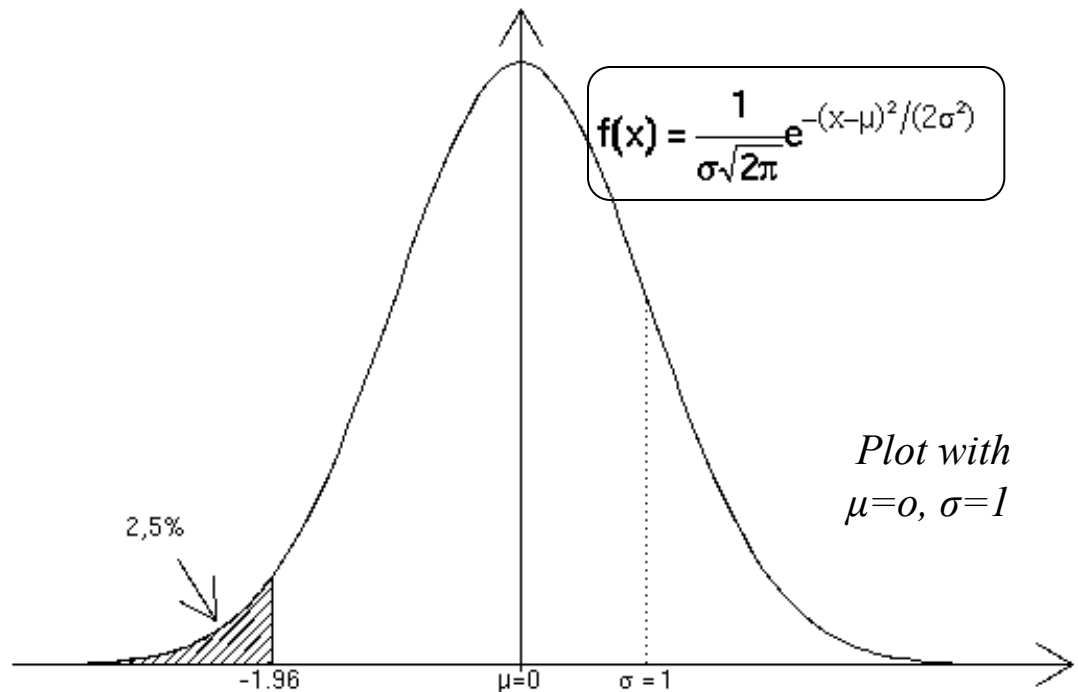


- Example of a *stationary point*: the gradient is zero
(while saddle points or local min/max are also stationary points)

Density (example)

- The density function of the normal random variable with mean 0 and variance 1 (called standard normal distribution, see the figure) and the analytical expression of the corresponding density in the generic case (*mean μ and variance σ^2*).

The probability density function (pdf) for a **normal distribution**:
“Gaussian function”



Next steps

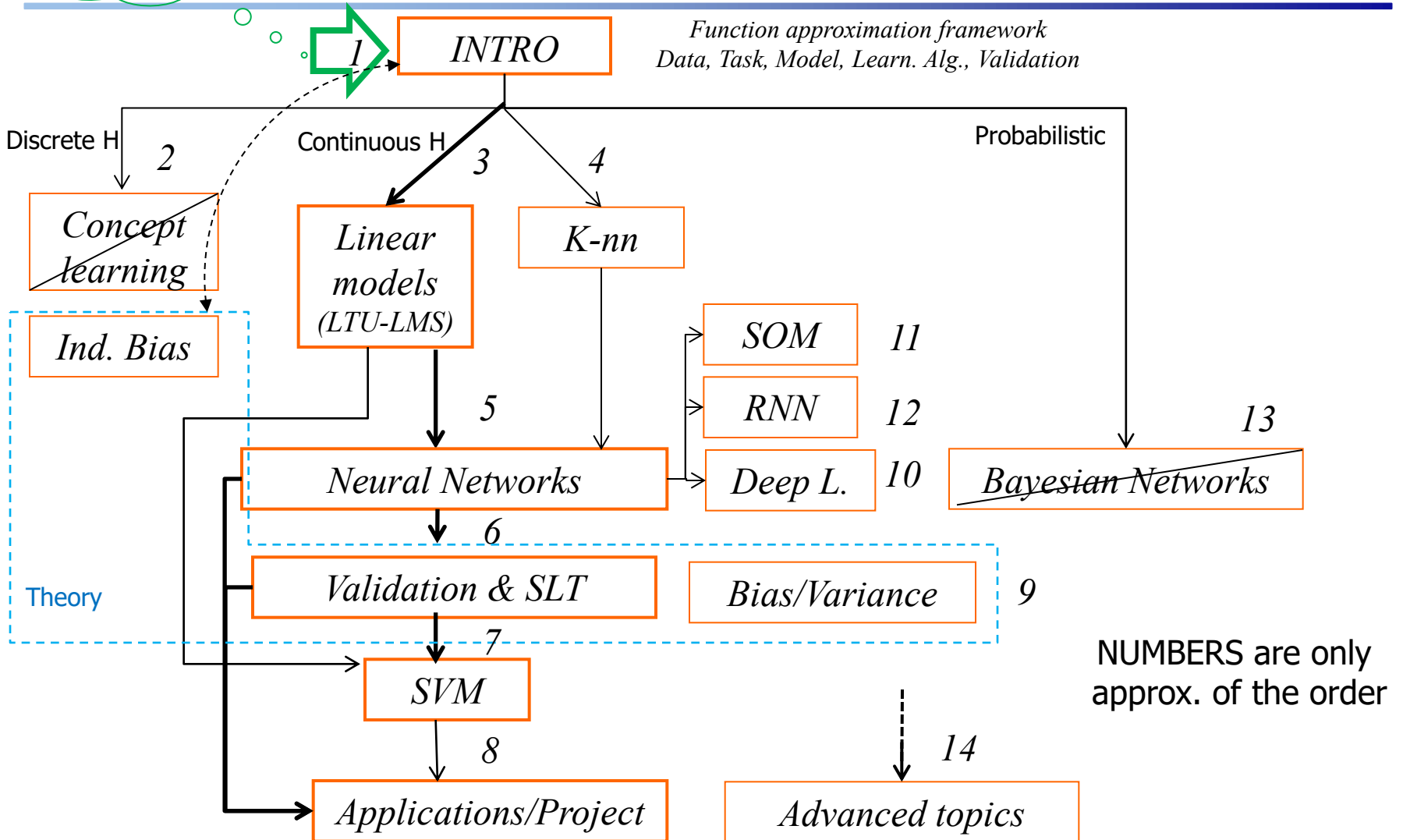


ML Course structure

- preview



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Summary of the Intro to ML

- Part I (now)
 - Motivations, contextualization in CS
 - Course info
- Part II (in Lect.s 2 and 3)
 - Utility of ML
 - Learning as function approximation (pilot example)
 - Design components of a ML system, including
 - Learning tasks
 - Hypotheses space (and first overview)
 - Inductive bias (examples in discrete hypotheses spaces)
 - Loss and learning tasks
 - Generalization (first part)
- Part III (in Lect. 4)
 - Generalization and Validation

Aim: *overview and terminology
before starting to study models and learning algorithms*

PROLOGUE END



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**Computational Intelligence & Machine
Learning Group**



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